

Jorge Novoa Contreras

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Research Interests

Nonlinear analysis, nonlinear differential equations, periodic boundary value problems, topological degree and continuation methods, monotone operator theory, variational methods.

Education

Universidad de Chile	Santiago, Chile
M.Sc. in Applied Mathematics	2025 – 2026
Final grade: 7.0/7.0 (highest possible grade)	
Thesis: <i>Periodic Solutions for Nonlinear Differential Equations in Banach Spaces with Generalized Variable-Exponent Operators</i>	
Advisor: Prof. Raúl Manásevich	
Thesis Committee: Prof. Jean Mawhin , Prof. Gabrielle Nornberg , and Prof. Alexander Quaas	
Mathematical Civil Engineering	2020 – 2025
Final grade: 7.0/7.0 (highest possible grade)	
B.Sc. in Engineering Sciences, major in Mathematics	2020 – 2024

Publications

1. Raúl Manásevich and **Jorge Novoa**, *Global continuum of solutions for systems of ODEs with periodic boundary conditions and generalized variable-exponent operators*, *Communications on Pure and Applied Analysis* **30** (2026), 151–166.

Research Experience

Universidad de Chile	Santiago, Chile
Research Intern	Dec. 2024 – Feb. 2025
• Conducted research on nonlinear differential equations, focusing on existence results for periodic systems involving ϕ-Laplacian-type and generalized variable-exponent operators, under the supervision of Prof. Raúl Manásevich.	

Teaching Experience

Universidad de Chile	Santiago, Chile
Assistant Instructor	2022 – 2026
• Conducted problem-solving sessions and tutoring in <i>Nonlinear Analysis</i>, <i>Ordinary Differential Equations</i>, <i>Multivariable Calculus</i>, and <i>Laboratory: Enhancing Your Mathematical Skills</i>.	
Teaching Assistant	2022 – 2025
• Graded quizzes, tests, and examinations in <i>Multivariable Calculus</i>, <i>Differential and Integral Calculus</i>, <i>Complex Variables and Special Functions</i>, <i>Linear Algebra</i>, and <i>Advanced Calculus and Applications</i>.	

Additional Experience

Center for Mathematical Modeling (CMM)

Santiago, Chile

Selection Reviewer, International Mathematical Modeling Challenge (IMMC)

Nov. 2023 – Dec. 2023

- Reviewed and evaluated mathematical modeling submissions as part of the national selection process for the teams representing Chile at the IMMC.

RialStat

Santiago, Chile

Professional Intern

Jan. 2023 – Feb. 2023; Dec. 2023 – Feb. 2024

- Worked with economic and financial datasets and supported statistical, macroeconomic, and machine-learning models for short-term forecasting and investment portfolio construction.

Languages

- **Spanish:** Native speaker.
- **English:** C1 level (IELTS Academic, overall band score: 7.5).